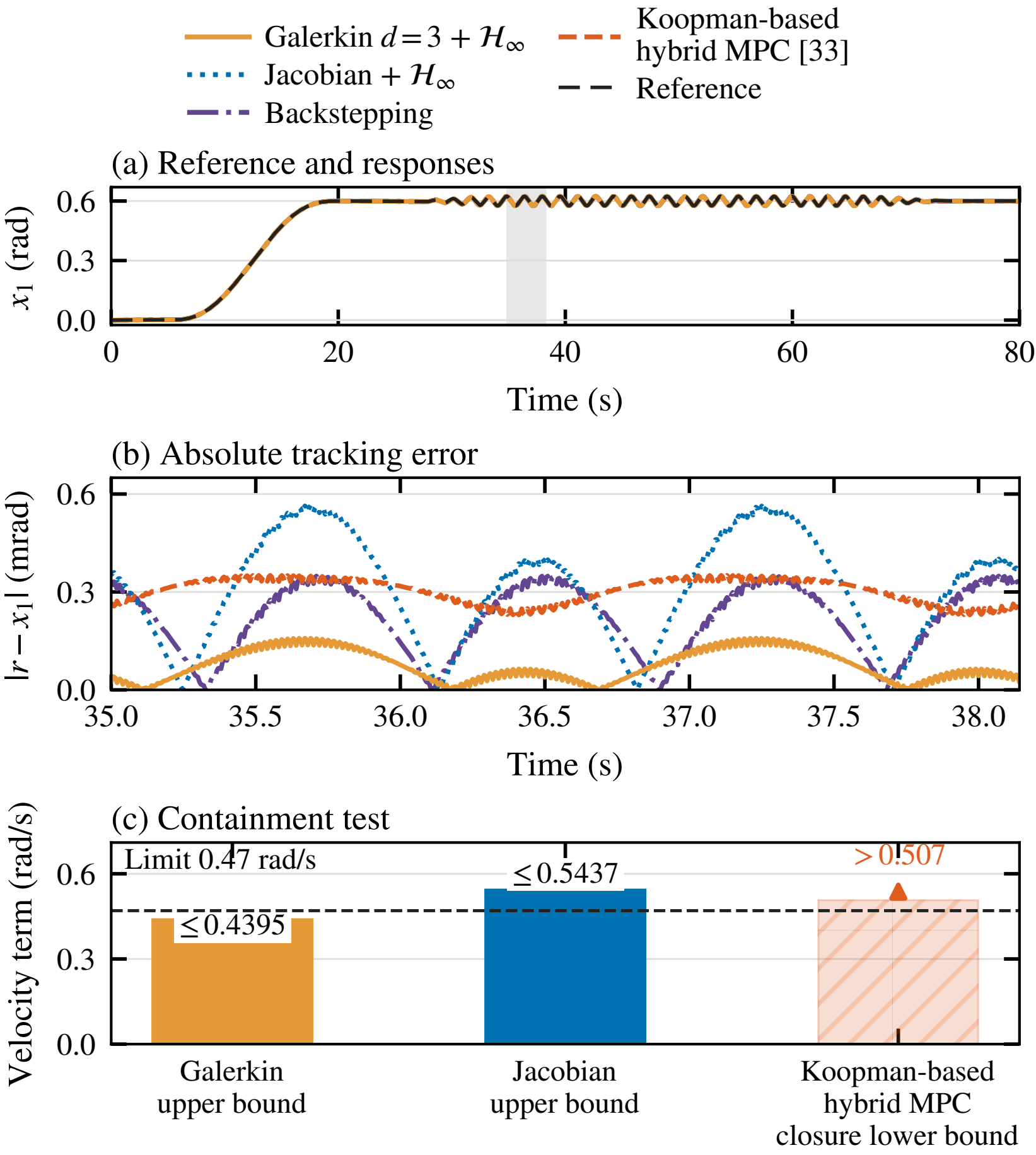
Galerkin $d=3$ + $\mathcal{H}_\infty$
Jacobian + $\mathcal{H}_\infty$
Backstepping
Koopman-based hybrid MPC [33]
Reference
(a) Reference and responses
x_1 (rad)
0.6
0.3
0.0
0
20
40
60
80
Time (s)
(b) Absolute tracking error
$|r - x_1|$ (mrad)
0.6
0.3
0.0
35.0
35.5
36.0
36.5
37.0
37.5
38.0
Time (s)
(c) Containment test
Velocity term (rad/s)
Limit 0.47 rad/s
0.6
0.3
0.0
≤0.4395
≤0.5437
>0.507
Galerkin upper bound
Jacobian upper bound
Koopman-based hybrid MPC closure lower bound